Steven Aarons

Queens, NY 11102
973-723-1144 | steven.aarons48@myhunter.cuny.edu | StevenAarons.com

EXPERIENCE

Ni-Meister Remote Sensing Research Lab – Hunter College New York, NY
Researcher October 2025 – Current

- Analyzed the relationship between ground level NO_2 from EPA AQS stations, satellite vertical column NO_2 from the new TEMPO Satellite, ERA5 meteorology, and traffic volume data in a NASA affiliated laboratory using Python.
- Developed an XGBoost machine learning model to predict ground level NO_2 using meteorology, transportation, and TEMPO input with a test dataset r^2 of 0.85.

NYC Department of City Planning – Environmental Assessment and Review Division New York, NY
Air Quality and Noise Intern June 2025 – August 2025

- Created 3 automated Air Quality HVAC screening tools that analyze site data, create symbolized map layers, and generate Excel reports, transforming a multi-day manual process into a 30-minute automated workflow.
- Developed two versions of a Construction Noise Analysis tool that processes multiple CSV files to calculate noise values, automatically outputting formatted Excel files and generating map layers.
- Developed a Python GUI application to convert Excel sheets to .csv files in a given output folder to accelerate exporting.
- Gained an understanding of the City Environmental Quality Review, CEQR, Technical Manual.

National Science Foundation – CUNY CREST Remote Sensing for the Environment New York, NY
Researcher May 2024 – June 2025

- Presented research through posters at the American Meteorological Society, American Geophysical Union, City Tech Summer Symposium conferences.
- Won first place in the Environmental Science Section at the Emerging Researchers National conference.
- Developed a data-driven precipitation prediction model using a dual U-net architecture with TensorFlow Keras on Google Collab, leveraging HRRR and RTMA Reanalysis data.
- Gained an understanding of Numerical Weather Prediction models and deep learning with no prior experience.
- Achieved a Receiver Operating Characteristic (ROC) Area Under the Curve (AUC) score of 0.92 for a binary rain/no-rain classification model with a 12-hour lead time.

Rockowitz Writing Center – Hunter College New York, NY
Front Desk Coordinator February 2024 – Current

- Managed scheduling for over 50 appointments per week, tracked attendance, and maximized booking capacity by filling open slots with wait-listed students.
- Created a Python GUI to parse and display data from the booking website and email clients who are running late through Azure Graph API, saving time and streamlining training for new coordinators.

PAST PROJECTS & SKILLS

NYS Mesonet Weather Station Point Interpolation Analysis StevenAarons.com
Gaza Damage Proxy Map using Sentinel-1 InSAR — 2024
GIS and Remote Sensing: ArcGIS Pro, QGIS, Google Earth Engine, ENVI
Programming Languages: Python (ArcPy, Pandas, Matplotlib, TensorFlow Keras, PyTorch) SQL (Postgre/PostGIS), R, JavaScript
Data Analysis and Presentation: STELLA Modeling Software, Excel, Word Processing, PowerPoint, Graphic Design
Language & Interests: Russian, Hiking, Landscape Architecture, Volunteer of Hart-to-Hart Community Garden

EDUCATION

CUNY Hunter College; Transferred from Vassar College
Bachelor of the Arts, Environmental Studies, Spatial Data Science Certificate New York, NY
Overall GPA: 3.77, Departmental Award Recipient, Dean's List, Conferral December 2025